Juan S John

Electronics & Computer Engineering student, VIT Chennai
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EDUCATION

B.Tech, Electronics & Computer Engineering (2023 - Present)
Vellore Institute of Technology, Chennai. Digital systems, signal processing, programming, embedded platforms, VLSI design.

Class XII, Science (PCM + CS)
Kerala.

EXPERIENCE

Summer Research Intern, Smart Healthcare (20 May 2026 - 31 Aug 2026)
Centre for Healthcare Advancement, Innovation and Research (CHAIR), VIT Chennai. Worked on noninvasive blood glucose estimation from ECG signals using a multi-expert SC-ResNet deep learning model.

PROJECTS

7:3 Compressor and 8x8 Wallace Tree Multiplier (team VLSI project)
Designed in Cadence Virtuoso; compared static CMOS and hybrid logic on power, delay, power-delay product and area.

Study Sync AI
Study companion web app for managing and organizing a personal study workflow (React).

Adaptive Audio Equalizer
5-band equalizer in MATLAB with a Java port, exploring context-aware DSP and real-time filtering.

Multi-Player Buzzer System
Embedded input system in C with strict prioritization logic and stable, low-latency response.

Interactive Online Draft Game
Browser-based strategic drafting game in HTML, CSS and JavaScript.

SKILLS

Languages: Java, C, Python, JavaScript, MATLAB
Web & tools: HTML/CSS, Git & GitHub, VS Code, Cadence Virtuoso
Engineering: Digital systems, signal processing, embedded platforms, circuit design, VLSI design
Concepts: DSA, OOP, ML basics, problem solving, prompt engineering